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Department of Crops, Horticulture, and Soils (CHS)

BOOK OF ABSTRACTS

Seed Potato Research Output

Publications, Innovations, & Climate–Smart Agricultural Technologies in Peer–Reviewed International Journals

FOCUS VALUE CHAIN
Seed Potato (Solanum tuberosum)

CORE THEMATIC AREA
Climate-Smart Agriculture (CSA-TIMPs)

TARGET REGION
Kenya

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1.0 Outreach & community engagement; Climate smart Agriculture Technologies Innovations and Management Practices (CSA-TIMPS)

1.1 Waaswa, A., Aniche, A., Nkurumwa, A.O., Kibe, A.M., and Kipkemoi, N. (2021).

W. Leal Filho et al. (eds.). **Exploring the Future of Agriculture under Climate Change: The Potentials of Climate-Smart Agriculture among Smallholder Farmers in Kenya.** ***Handbook of Nature-Based Solutions to Mitigation and Adaptation to Climate Change*****.**

Springer; https://doi.org/10.1007/978-3-030-98067-2_146-1

Abstract

Climate variability, characterized by low precipitation, flooding, high temperatures, prolonged sunshine, and delayed rainfall, has threatened agricultural productivity, leading to food insecurity and chronic poverty, especially among resource-constrained farmers. Several climate-smart agriculture (CSA) strategies have been developed to address these challenges. Based on the CSA practices diffused among Kenyan farmers, this study tested a hypothesis that adopting CSA is a significant strategy for climate change adaptation. A cross-sectional survey was used to collect data from a random sample of 120 smallholder potato farmers in Gilgil Sub-County of Kenya. Data was collected using a structured researcher administered questionnaire, and the Statistical Package for the Social Sciences version 28 was used for data analysis. The percentages and frequencies for the CSA adoption rates and potentials of different CSAPs were computed using descriptive analysis. A binary regression analysis revealed that CSA is a significant climate change adaptation strategy (Wald $\chi 2$ ¼ 49.417, df ¼ 1, $p < 0.001$). Further, implementing CSA increased the farmers' chances to adapt to climate change by 19 times more than non-adopters. Explicitly, this study revealed that CSA significantly contributed to farmers' adaptation potentials through reducing soil erosion, increasing livelihood diversification, reducing input expenses, and increasing product quality. To exploit the benefits of CSA, this study recommends using experiential learning approaches, establishing linkages between

research institutions, and capitalizing on progressive farmers to hasten the acceptance of CSA.

Conclusion and Recommendations

The study revealed an average **of 64.56% of the farmers adopted CSA practices** and synthetic fertilizers were the most adopted CSA, with the mini-tubers being the least adopted. This study tested the hypothesis that adopting **CSA is a significant strategy for climate change adaptation, and results showed that CSA increased the farmers' chances to adapt to climate change by 19 times more than non-adopters.**

Specifically, CSA significantly contributed to farmers' adaptation potentials by;

i. **reducing soil erosion,**
ii. **increasing livelihood diversification,**
iii. **reducing input expenses,**
iv. **increasing product quality.**

The study's main findings demonstrate that smallholder farmers who adopted CSA reaped benefits that would translate into a ripple of positive impacts. For example, reducing soil erosion **preserves soil fertility; increasing livelihood diversification spreads the risk associated with income generation and protects them against economic depression; reducing input expenses motivates farmers to adopt more efficient and sustainable farming practices that minimize environmental impacts and promote eco-friendly agriculture; and increasing product quality** results in fewer rejected or unsellable items due to defects or subpar characteristics. **This reduces waste along the supply chain, minimizing losses for farmers and promoting sustainable practices.** To exploit the benefits of CSA, this study recommends using experiential learning approaches, establishing linkages between research institutions, and capitalizing on progressive farmers to hasten the acceptance of CSA. This would allow farmers **to actively engage with new CSA practices and technologies, leading to better understanding and adoption.** Building strong connections between research institutions and farmers would facilitate the **exchange of information and expertise, creating a dynamic feedback loop that fosters the acceptance of CSA**. Lastly, progressive farmers can serve as role models and extension agents within their communities.

They can share their experiences, demonstrate the benefits of CSA, and provide practical guidance to their fellow farmers.

1.2 Communicating climate change adaptation strategies: climate-smart agriculture information dissemination pathways among smallholder potato farmers in Gilgil Sub-County, Kenya. *Heliyon*. https://doi.org/10.1016/j.heliyon.2021.e07873/

Waaswa, A., Nkurumwa, A.O., Kibe, A.M., and Kipkemoi, N. (2021).

Abstract

Proven and sustainable practices like climate-smart agricultural practices (CSAPs) need to be prioritized and promoted for uptake especially by the farmers to achieve sustainable development. These are capable of contributing to the realization of sustainable development goals through averting food and nutritional insecurity, increasing and sustaining yields that translate into increased incomes and later reduced poverty. This is because CSAPs enable farmers to adapt and mitigate climate change effects. However, due to inappropriate communication of CSAPs to the farmers, to date, some farmers still see no escape route from the frightening effects of climate change and they are currently adopting a rather fatalistic attitude. This study investigated the information dissemination pathways used by different categories of smallholder potato farmers for and practice of CSAPs. It found a difference between information sources and practice of CSAPs at a 5% level of significance ($\chi 2$ ¼ 100.12139, df ¼ 2, $p < 0.05$, Cramer's V ¼ 1.0), and a difference in the use of the three information dissemination pathways between men and women at a 5% level of significance ($\chi 2$ ¼ 6.05949, df ¼ 2, $p < 0.05$, Cramer's V ¼ 0.17406). **The three information dissemination pathways included media, neighbors and friends, and extension officers**. Generally, farmers were aware and practiced the CSAPs investigated in this study except for irrigation with high awareness yet with low uptake percentage and potato seedlings and minitubers both with low awareness and practice respectively. This study recommended mainstreaming of CSAPs information.

Conclusions and policy implications

This study investigated the information dissemination pathways used by different categories of smallholder potato farmers for the awareness and practice of CSAPs. It found a difference

between information sources and practice of CSAPs and a difference in the use of the three information dissemination pathways between men and women. Extension officers scored higher for both men and women, though men sourced more information from them than women did. However, women sourced more information from neighbours and friends than men did, while media was mostly used by men. Averagely, extension officers were the most used source of awareness followed by neighbours and friends, and media respectively. Additionally, women were more aware of CSAPs than men were. Broadly, farmers were aware of the CSAPs investigated in this study except for potato seedlings and minitubers that yielded low awareness percentages. Additionally, a relatively large number of farmers practiced most CSAPs, which they indicated they were aware of, except for irrigation that scored high on awareness yet with a low score on practice. Following these conclusions, there is a need to appreciate differences in the use of different information dissemination pathways by men and women for the awareness and practice of CSAPs. In addition, broad consideration of different information sources with the potential to result in bumper uptake of CSAPs would go a long way in ameliorating climate change impacts among farmers. Further, this study recommends mainstreaming of CSAPs information

1.3 Adapting agriculture to climate change: Institutional determinants of adoption of climate-smart agriculture among smallholder farmers in Kenya

Waaswa, A., Nkurumwa, A.O., Kibe, A.M., and Kipkemoi, N. (2024) . ***Cogent Food & Agriculture*****, 10:1, https://doi.org/10.1080/23311932.2023.2294547/**

Abstract

Climate-smart agriculture (CSA) adoption rates have remained low in Kenya despite being promoted by the Kenyan government and its development partners. Analyzing institutional factors could help inform efforts to mitigate potential climate adaptation mal-actions in Kenya and other contexts. This study examined the relationship between institutional factors and CSA adoption among smallholder potato farmers in Gilgil Sub County of Nakuru County, Kenya. The institutional factors selected for this study included access to credit, training on CSA, non-governmental organization (NGO) support, and farmers' group membership. A binary logistic regression analysis unveiled that smallholder potato farmers' adoption of CSA was positively related to institutional factors, which was statistically significant at a 5% significance level (χ= 10.219, df = 4, $p < 0.05$). However, only access to credit was positive and statistically significant at a 5% significance level (Wald $\chi 2 = 4.524$, df = 1, $p < 0.05$) among the four explanatory variables included in the binary logistic regression model. Therefore, warranting **access to credit is deemed to produce favorable requisites for adopting farming methods** befitted to adapting agriculture to the effects of climate change. One way of warranting access to credit could be mobilizing farmers to join groups that serve as Savings and Credit Cooperatives (SACCOs) and Village Savings and Lending Associations (VSLAs) because farmers that join SACCOs or VSLAs have access to loans that may otherwise be challenging to obtain from conventional financial institutions.

Conclusions and policy implication

This study found a positive and significant relationship between institution factors, credit access, particularly, and adoption of CSA. Access to credit is therefore thought to result in favorable conditions for implementing farming practices appropriate for adjusting agriculture to the effects of climate change. **One way to ensure access to credit should be by mobilizing farmers to join groups that serve as SACCOs and Village Savings and Lending Associations (VSLAs).** This is because farmers who join SACCOs or VSLAs have access to loans that may otherwise be challenging to obtain from conventional financial institutions.

These organizations enable members to pool their savings and lend money to one another depending on the need and ability to repay, and farmers can lessen their reliance on moneylenders, who frequently demand exorbitant interest rates. SACCOs and VSLAs would grant farmers access to farming resources like tools, equipment, and knowledge and provide financial help through cooperative sharing, which may increase **adaptation potential and agricultural production. This study calls for increased efforts to facilitate smallholder farmers' access to credit since** it is critical to fostering CSA adoption.

2.0 ENTREPRENEURSHIP (SOCIAL-ECONOMIC ISSUES)

2.1 Understanding the socioeconomic determinants of adoption of climate-smart agricultural practices among smallholder potato farmers in Gilgil Sub-County, Kenya

Waaswa, A., Aniche, A., Nkurumwa, A.O., Kibe, A.M., and Kipkemoi, N. (2021). Understanding the socioeconomic determinants of adoption of climate-smart agricultural practices among smallholder potato farmers in Gilgil Sub-County, Kenya. *Discover Sustainability*, 2:41. https://doi.org/10.1007/s43621-021-00050-x/

Abstract

Besides climate-smart agriculture's (CSA) potential to meet the world's increasing food demands in the face of climate variability through sustainably increasing food production, its acceptance among farmers is still low. This could be partly because of limited insight into the contextual underpinnings of its uptake. Therefore, the purpose of this study was to establish the relationship between selected socioeconomic factors and the adoption of CSA in Gilgil Sub-County. This study's results were attained from a binary logistic regression model, using a sample of 120 smallholder potato farmers in two wards of Gilgil Sub-County of Nakuru County, Kenya. An analysis of the five hypothesized explanatory socioeconomic variables contained in the model disclosed that a relationship between socioeconomic factors and adoption of CSAPs was statistically significant at a 5% level of significance ($\chi 2 = 17.966$, df = **5, $p < 0.05$). It further revealed that only two variables had a significant relationship with the adoption of CSAPs. Among these, included gender which was negative and statistically significant at a 5% level of significance (Wald $\chi 2 = 6.701$, df = 1, $p < 0.05$) and annual farm income, which was positive and statistically significant at a 5% level of significance (Wald $\chi 2$**

= 8.402, df = 1, $p < 0.05$). Therefore, securing access to vital resources for women farmers is indispensable to enhance their capacity and compliance to adjust production methods in response to climate change. Facilitating increased farm output and income among the farmers is greatly recommended.

Conclusions and policy implication

This study used a binary logistic regression model to establish the relationship between selected socioeconomic factors and the adoption of CSAPs among smallholder potato farmers in Gilgil Sub-County, Kenya. An analysis of the five hypothesized explanatory socioeconomic variables in the model disclosed that a relationship between socioeconomic factors and the adoption of CSAPs was statistically significant. **It further revealed that only two variables had a**

significant relationship with the adoption of CSAPs. Among these, included gender which was negative and statistically significant, and annual farm income that was positive and statistically significant. This study's empirical results present valuable advice for policymakers. To begin with, the results in this article gives proof that annual farm income and gender are pivotal socioeconomic factors in affecting the possibilities of farmers to adopt CSAPs. Therefore, policies that support farmers to increase farm earnings should be prioritized. These will help create favourable requisites for the scaling and adoption of farming methods befitted to adapting agriculture to the effects of climate change. Additionally, facilitating increased farm output and increased incomes among the farmers is greatly recommended. Besides, gender turned out to be negative and significant, implying that women are pivotal in shaping the likelihood of practicing CSAPs in the face of climatic changes. Thus, agricultural technologies should not be developed without consideration of conditions that favour access to such technologies by women. Consequently, securing access to vital resources for women farmers is indispensable to enhance their capacity and compliance to adjust production methods in response to climate change. Nevertheless, like other studies, this is not without imperfections. Principally the examination was restricted to cross-sectional data. This constrains the observation of long-term transitions and changes in farmers' decisions to adopt CSAPs. This flaw should be considered while assessing this study's deductions and forms a potential area for further research.

2.2 Factors Influencing Choice of Differentiation Strategy among Small-Scale Potato Enterprises in Nakuru County, Kenya

Agbolosoo,J.A., Kariuki, I.M., Bett, H.K. & Nkurumwa, A.O. (2021). Factors Influencing Choice of Differentiation Strategy among Small-Scale Potato Enterprises in Nakuru County, Kenya. ***African Journal of Pure and Applied Science Research*****, 2:1**

Abstract

Potato crop plays a crucial role in national food nutrition and ensures food security that alleviates poverty among small-scale potato farmers in Kenya. Potato farming is characterized by several constraints leading to declining production and yields in Kenya. This paper determined factors influencing the choice of differentiation strategy among small-scale potato enterprises in Nakuru County, Kenya. The study used a descriptive research design to produce statistical information about factors influencing differentiation strategy among small-scale potato enterprises. A semi-structured questionnaire was used to elicit information from a sample of 267 respondents in Molo Sub County where the primary data collected was analyzed using the probit regression model. The study findings indicate that age, education level, farmer group, access to storage, proactiveness behaviour, and cosmopoliteness behaviour influenced the choice of differentiation strategy among small-scale potato enterprises. It was concluded that participation in farmer groups and access to a storage facility and entrepreneurial behaviour skills drive small-scale potato farmers to build and maintain a sustainable competitive advantage in the small-scale potato enterprises in Nakuru County. The government needs to provide technical assistance, credit support, and

provide storage facilities for potato farmers to improve competitiveness in the small-scale potato enterprises.

Conclusion and Recommendations

The study concludes that age and education level of small-scale farmers, membership to farmer group, and access to a storage facility, proactiveness behaviour, and cosmopoliteness behaviour influence choice of differentiation strategy among small-scale potato enterprises in Molo Sub County, Kenya. It is recommended that small-scale potato farmers need to participate more in farmer groups to receive training on the adoption of improved seed varieties and production techniques in the small-scale potato enterprises. The government in partnership with the private

sector should provide technical assistance, credit support, and storage facilities for potato farmers to enhance the competitive advantage of small-scale potato enterprises.

2.3 Role of Networking Capability, Socio-Economic and Institutional Characteristics on Adoption Tendencies of Clean Seed Potato Agri-Enterprises in Central Rift Valley, Kenya

Ong'ayo, M.J., Gido, E.O., Ayuya, O.I Mwangi, M., and Kibe, A.M. (2020). Role of Networking Capability, Socio-Economic and Institutional Characteristics on Adoption Tendencies of Clean Seed Potato Agri-Enterprises in Central Rift Valley, Kenya. ***African Crop Science Journal*****, 28, 131-144.**

Abstract

Even though quality of seed is a major yield determinant in potato (*Solanum tuberosum* L.) production and global food security, inadequate availability of and access to high quality (certified) seed is a major challenge to potato producers in Sub-Saharan Africa. The objective of this study was to examine farmer's adoption tendency towards decentralised *clean seed* potato multiplication agri-enterprise (CSPMAE) in Central Rift Valley, Kenya, which aids in implementation of stage matched entrepreneurial interventions. A cross-sectional survey was conducted using a semi-structured questionnaire **on 54 seed potato producers and 192 non-seed potato producers,** through multistage sampling. Ordered logistic regression analysis, based on the trans-theoretical model, **revealed that household head, partner knowledge, gender, land size allocated for potato production, level of education, ownership of transport and/or storage facilities, frequency of interaction with certified seed potato producers and agricultural extension officers influenced adoption tendencies of CSPMAE, positively and significantly**Seed potato value chain stakeholder platforms, development workers and donors should prioritise their support for *clean seed* agri-enterprises to farmers with such traits. This is likely to lead to increased supply of access to *clean seed,* thereby improving potato yields in Kenya. **Conclusion**

It is clear from this study that among the networking capability constructs, partner knowledge of the household head and their ability to build new relations play a positive role in adoption tendency of CSPMAE in the central Rift Valley of Kenya (Nakuru). Gender of the household head, land allocated to potato and level of education of the household head are among the socio-economic factors that positively influence the uptake stages for decentralised CSPMAE in this region. Major institutional factors include ownership of transport, ownership of storage facility, interaction with extension services and interaction with certified seed potato producer; all of which had effects on adoption tendencies of decentralised CSPMAE in the study area.

3.0 Early seed Generation; Laboratory Cultures

3.1 Effects of Sucrose and Gibberellic Acid on Growth and Survival of Local Potato (Solanum tuberosum L.) Varieties in vitro in Kenya

Makau, F.M., Mwangi, M., Oyoo, M.E., Kibe, M.A. & Oggema, J. (2022). Effects of Sucrose and Gibberellic Acid on Growth and Survival of Local Potato (Solanum tuberosum L.) Varieties in vitro in Kenya. ***European Journal of Biology and Biotechnology,*****3: 4. https://10.24018/ejbio.2022.3.4.372/**

Abstract

Tissue culture techniques' have become useful technologies for producing disease & pest free seed potato (Solanum tuberosum L.) in the developed world. However, these techniques have yet to be standardized for locally produced potato varieties in Kenya. Developing countries can also use these innovations for rapid multiplication of popular local seed material through rooted apical cuttings generated from either plantlets or micro tubers. In vitro experiments were therefore conducted to determine the optimum concentration of sucrose and gibberellic acid for growth and survival of local potato varieties, namely, Shangi, Unica and Wanjiku. The explants were cultured in Murashige and Skoog (MS) media supplemented with sucrose at a concentration of 20, 30 and 40 gL-1, while gibberellic acid was applied at a concentration of 0.2, 0.5 and 1.0 mgL-1. The study was laid out in a completely randomized design (CRD).

i. **MS Medium with sucrose 40 gL-1and gibberellic acid 0.5 mgL-1 significantly enhanced shoot length, with the longest shoot (10.3 cm) being recorded for Wanjiku.**
ii. **The same treatment also gave the highest plant survival of 90%. Murashige and Skoog media, added with 0.5 mgL-1 gibberellic acid along with 40 gL-1 sucrose is recommended for generating wanjiku, unica and shangi apical rooted cuttings** because it gave the best improvement of in-vitro clonal growth.

Conclusions

Using MS Medium in growth of in vitro and sucrose as a carbon source at the rate of 40 gL-1 gave better growth in terms of leaves number, number of shoots, shoot length and the number of roots. The supply of gibberellic acid (GA3) at a concentration of 0.5 mgL-1 combined with sucrose at a concentration of 40 gL-1 significantly recorded a better shoot length, the number

of shoots, number of roots, number of leaves, root initiation, shoots fresh weight and shoots dry weight on local potato varieties cultured in vitro. Potato varieties responds differently to different treatment levels. **These results showed that 0.5 mgL-1 GA3 along with 40 gL-1 Sucrose had better effect on survival and improvement of shoot growth.**

4.0 Protected (Greenhouse/ Screenhouse) Cultures

4.1 Influence of Hydroponics Nutrient Solution on Quality of Selected Varieties of Potato Minitubers

Wambugu, W.C.; Kibe, A.M.;Opiyo, A.M.; Githeng'u, S.; Odong, T. Influence of Hydroponics Nutrient Solution on Quality of Selected Varieties of Potato Minitubers. Orticulturae, 10, 126. https://doi.org/10.3390/horticulturae10020126/

Abstract

Addressing poor seed quality is pivotal for increased potato yields in Kenya. For this to be realized there is a need for nutrient optimization in the hydroponic system. The objective of this study was to examine the effects of nutrient stock solution concentrations on the quality of minitubers produced under a hydroponic system. Two greenhouse experiments were set up at Egerton University, Kenya in 2022. The treatments included three nutrient solution concentrations: 75% (N75), 100% (N100) and 125% (N125) and four potato varieties (Wanjiku, Unica, Shangi and Nyota) grown in a cocopeat substrate hydroponic system. **The results indicated that the application of N125 produced minitubers that had significantly higher specific gravity, dry matter, starch, ash and sugar content.** Crude protein and phosphorus did not differ significantly with the application of varying nutrient concentrations. The varieties did not differ significantly in the quality parameters except for **total sugars where Unica was significantly different from Nyota and Wanjiku while Shangi did not differ from all varieties.** Therefore, it will be advisable to apply 125% of the ADC-Molo recommended nutrient stock formulation which should be considered as an effective method of increasing minitubers quality under a hydroponic system

Conclusion

The results of the current study showed that increasing the nutrient stock solution concentrations **from 75% to 125% of the ADC Molo nutrient formulation increased minituber quality parameters, i.e., starch, sugars, dry matter, crude protein, phosphorus, specific gravity and ash content which could contribute to quality potato production.** Considering the superiority of the varieties, all varieties did not differ significantly in all

parameters

apart from sugars where Unica had high content as compared to Nyota and Wanjiku. Understanding the role of adequate nutrient supply in quality seed production should be taken into account. Therefore, it is recommended to apply 125% of ADC-Molo nutrient formulation for optimal minitubers quality. However, it is crucial to acknowledge that the higher nutrient concentration may not necessarily translate into lower production costs. Further research is recommended to evaluate the growth and yield performance of minitubers produced under varying nutrient stock solution concentrations.

4.2 Effects of Media Type on Growth and Yield of Potato (Solanum tuberosum L.) Varieties Apical Rooted Cuttings in Kenya

Makau, F.M., Mwangi, M., Oyoo, M.E., & Kibe, M.A. (2023). Effects of Media Type on Growth and Yield of Potato (Solanum tuberosum L.) Varieties Apical Rooted Cuttings in Kenya. *European Journal of Science, Innovation and Technology*, 3, 3. www.ejsit-journal.com/

Abstract

To address soil-borne virus diseases; a soilless media has been introduced for producing healthy seed tuber based on in vitro rapid multiplication of virus-free planting material. The objective of this study was to determine the effect of different media on the growth of potato variety using apical rooted cuttings. A greenhouse study was carried out where five media types; coco-peat + perlite, sand, coco-peat + pumice, coco-peat + vermiculite, and soil were tested along with three local potato varieties (Shangi, Unica and Wanjiku). The study was a 3×5 factorial arrangement laid out in a randomized complete block design (RCBD). Data collected was subjected to a general linear model (GLM) to partition the variance component using SAS software version 9.0 and means separated using Tukey's Honestly Significant Difference Test (HSD) at $P \leq 0.05$. The interaction between varieties and different media had a significant effect on the number of tubers per plant**. Shangi registered the highest number of mini tubers per plant (21tubers) on coco-peat + perlite which was not significant different from that of coco-peat +pumice mixture (19 tubers)**. Plant height of 53cm was recorded under cocopeat + perlite mixture. Highest shoot fresh weight was recorded on Shangi (24.86g/plantlet) cultivated on coco-peat + pumice mixture, followed by Unica and Wanjiku (24.64 and 21.17g/plantlet respectively) cultivated on coco-peat + perlite mixture. **Unica produced the**

highest yield (64.8g/plantlet) in coco-peat + pumice, followed by Wanjiku (59.8g/plantlet) and Shangi (40.8g/plantlet) in coco-peat + perlite. This study recommends that for farmers to overcome the spread of pests and diseases, use of coco-peat + perlite and coco-peat+pumice as a soilless media should be promoted to enhanced seed potato growth and tuber formation.

Conclusion

The performance of potato varieties varies with propagating media. Coco-peat + perlite was found to be superior in supporting plant growth and tuber formation, followed by cocopeat + pumice. The study has provided valuable information on the media type and the role it plays in increasing the productivity of seed tuber in perspective of commonly used local varieties in Kenya

4.3 Effect of nutrient solution concentrations on the growth and yield of potato (*Solanum tuberosum* L.) varieties grown from apical rooted cutting in a hydroponic system

Wambugu, W. C., Opiyo, A. M., & Kibe, A. M. (2023). Effect of nutrient solution concentrations on the growth and yield of potato (*Solanum tuberosum* L.) varieties grown from apical rooted cutting in a hydroponic system. *Archives of Agriculture and Environmental Science*, *8*(3), 325-332, https://dx.doi.org/10.26832/24566632.2023.080308/

Abstract

This study evaluated the effects of nutrient stock solution concentrations on the growth and yield of potato varieties grown from apical rooted cuttings (ARCs). A greenhouse experiment was conducted at the Climate and Water Smart Agriculture Center at Egerton University, Kenya. The experiment was laid out in a split-plot arrangement in randomized complete block design, where the main plot comprised three nutrient concentrations, i.e., 75% (N75), 100% (N100) and 125% (N125) of the ADC-Molo' nutrient formulation. The subplots were allocated to four potato varieties (Shangi, Wanjiku, Nyota and Unica). The results showed that there were no significant (p<0.05) interaction effects of the nutrient stock solution concentrations application rates on the growth attributes of ARCs. The main effects of N125 gave the tallest plants (32.29cm) at 60 days after planting (DAP), highest normalised difference vegetation index (NDVI) (0.60) at 75 DAP, plant survival rate (82.15%) at 75 DAP, and fresh weight (79.04g) and dry matter (31.26%) of aboveground biomass (AGB). Nyota variety produced taller plants (26.90cm) at 60 DAP, gave higher NDVI values (0.53) at 75 DAP, and higher fresh weight (64.87g) and dry matter (27.60%) of the AGB. Significant (p<0.05) interactions were observed in the yield parameters. The interaction between N125 and Nyota (11.33) and Wanjiku (10.67) gave the highest number of minitubers, the highest yields were obtained between the interaction of N125 and Unica (16.38t/ha). Therefore, to achieve high growth and yields of ARCs under hydroponic system, seed potato producers should use 125% of the ADC Molo nutrient formulation.

Conclusion

In this study, the application of nutrient stock solution concen-trations (NSSC) significantly affected growth and yield of ARCs. Increasing NSSC from 75% to 125% increased growth and

yield parameters of the apical rooted cuttings. Potato varieties re-sponded differently in growth and yields under the hydroponic system. Nyota and Wanjiku varieties gave higher growth rateand produced higher numbers of minitubers while Unica gave the highest yields (t/ha). This study therefore recommends that the application of 125% of the "ADC-Molo nutrient formulation" for optimum growth and yield of potato varieties grown from apical rooted cuttings under a hydroponic system maybe the appropriate rate. Further studies on the effects of individual nutrient elements on the growth and yield of potato apical root-ed cuttings are recommended.

5. Field Production Of potatoes

5.1 Water use efficiency and nitrogen use efficiency of apical rooted cuttings of potato grown in a Mollic Andosol

Satognon, F,. Lelei, J.J,. Owido, S.F.O. (2022). Water use efficiency and nitrogen use efficiency of apical rooted cuttings of potato grown in a Mollic Andosol. ***Urban Agriculture & Regional Food Systems*****. https://doi.org/10.1002/uar2.20026**

Abstract

Water and nitrogen (N) are key interacting factors that control plant growth. The objective of the study was to determine the effect of N rates and drip irrigation regimes on water use efficiency (WUE) and N use efficiency (NUE) of apical rooted cuttings of potato (*Solanum tuberosum* L.) grown in a mollic Andosol. The treatments comprised four drip irrigation regimes of 50%, 75%, 85% and 100% of the crop evapotranspiration (ETC), where ETC100% was irrigated according to the soil water depletion 2 d after full irrigation and four N rates: 0 (N0), 60 (N1), 90 (N2) and 130 kg N ha−1 (N3). The results showed that potato tuber yield, WUE, and NUE were significantly ($P < .001$) affected by irrigation regimes, N rates, and the interaction of both factors. **The highest potato tuber yield, 58.29 Mg ha−1, was obtained under ETC 100% with N3, whereas the highest WUE, 17.5 kg m−3, was found under ETC50% with N3. The apical rooted cuttings of potato grown under ETC100% with N2 produced the maximum NUE, 208.30 kg kg−1.** The correlation coefficient ($r = 0.31$; $P <$ .01) obtained between WUE and NUE was weak. It was also observed that an increase in irrigation amount decreased WUE and increased NUE, whereas a high N dosage increased WUE and decreased NUE of potato grown in a mollic Andosol. This study suggests that if the farmers aim to maximize NUE of apical rooted cuttings in a mollic Andosol, then they will do so at the expense of potato WUE, and yield will be looked at as a secondary factor.

Conclusion

Careful N management in terms of time, rate, and the method of N application is essential to achieve high potato yield while saving water and reducing N losses and considering the high cost of irrigation water and fertilizer. This study shows that potato tuber yield obtained from apical rooted cuttings increased with an increment of irrigation amount and N rate in a mollic Andosol. Maximum irrigation amount reduced WUE and increased NUE, whereas high N dosage increased WUE and decreased NUE of apical rooted cuttings of potato grown in a mollic

Andosol. Therefore, if maximizing NUE of apical rooted cuttings of potato in mollic Andosols is the objective of the farmers or researchers, then it should be at the expense of potato NUE, and yield should be looked at as the secondary target. For an optimal NUE of apical rooted cuttings of potato in a mollic Andosol, we recommend farmers to apply ~90 kg N ha^{-1} with an irrigation regime of ETC85%.

5.2 Performance of apical rooted cuttings of potato grown in Mollic Andosols under different nitrogen fertilization and irrigation regimes

Satognon, F,. Lelei, J.J,. Owido, S.F.O. (2021). Performance of apical rooted cuttings of potato grown in Mollic Andosols under different nitrogen fertilization and irrigation regimes. *Heliyon.* https://doi.org/10.1016/j.heliyon.2021.e07999

Abstract

High potato productivity (*Solanum tuberosum* L) is generally influenced by several factors, including water and nitrogen (N), and potato requirement for these factors varies depending on the soil type and potato variety. This research aimed to determine the performance of apical rooted cuttings of potato grown in Mollic Andosols under different nitrogen fertilization and irrigation regimes. The treatments comprised 4 irrigation regimes of 100%, 85%, 75% and 50% of the crop evapotranspiration (ETC), where ETC100% was irrigated based on water depletion in the root zone two days after full irrigation, and 4 nitrogen rates of 0 (N0), 60 (N1), 90 (N2) and 130 kg.ha-1 (N3) applied in splits at 10 (40%), 30 (40%) and 50 (20%) days after planting. The results revealed that the water demand for apical rooted cuttings of potato (ETa) was on average 201.4, 302.1, 342.4 and 402.8 mm under ETC50%, ETC75%, ETC85% and ETC100%, respectively. It was observed that plant height and number of branches significantly ($P < 0.001$) varied under different N rates with the highest plant height (92.67 cm) and number of branches per potato plant (17) achieved when applying N3. Potato grown under full irrigation (ETC100%) with N3 produced the highest total potato tuber yield (58.28 t.ha-1) and marketable tuber yield (54.21 t.ha-1). The number of tubers per plant statistically reduced as the N deficit increased, with the maximum tuber number, 23, achieved under N3. It was observed that a significant Pearson correlation ($r = 0.7$***) existed between tuber number and total tuber yield. The maximum harvest index (HI), 57.12 %, was obtained under ETC50% with N3, while the highest tuber dry matter, 30 %, was observed under N3. To achieve a high

tuber yield from apical rooted cuttings of potato in Mollic Andosols, this study recommends an irrigation regime of ETC100% and a nitrogen rate of 130 kg.ha-1.

Conclusion

Irrigation and N fertilization are the key factors in potato production. This study indicated that the difference in soil moisture content under potato production in Mollic Andosols was low during the first four weeks. The cumulative actual crop evapotranspiration (ETa) estimated in this study was on average 201.4, 302.1, 342.4 and 402.8 mm under ETC50%, ETC75%, ETC85% and ETC100%, respectively. Potato plant height, number of branches per plant, tuber number per plant and DM were generally responsive for N rate, while total tuber yield, marketable tuber and HI were more responsive to the interaction of both factors than a single factor. This study recommends an irrigation regime of ETC100% and N fertilizer rate of 130 kg.ha^{-1} in three split applications at 10 (40%), 30 (40%) and 50 (20%) days after planting for a maximum potato yield in Mollic Andosols in Kenya when planting apical rooted cuttings.

5.3 Effects of Calcium Nitrate Levels and Soaking Durations on Coco peat Nutrient Content Journal: Journal of Agricultural Chemistry and Environment

Gbollie, S.N., Mwonga, S.M. and Kibe, A.M. (2021) Effects of Calcium Nitrate Levels and Soaking Durations on Cocopeat Nutrient Content. ***Journal of Agricultural Chemistry and Environment,*** **10, 372-388. https://doi.org/10.4236/jacen.2021.103024 Abstract**

Cocopeat, a by-product of the coconut (Cocos nucifera L.), is an important soilless media that contains high potassium (K), sodium (Na), and electrical conductivity (EC) depending on its source. Methods for extracting these elements and thus lowering EC are yet to be standardized. This study was therefore carried out to investigate two extraction methods of these elements in cocopeat. A greenhouse pot experiment was carried out at the Climate and Water Smart Agriculture Centre of Egerton University, Kenya. It was laid out in a 5 × 4 factorial completely randomized design. Five soaking durations (12, 24, 36, 48, and 72 hours) and four calcium nitrate (Ca(NO3)2) levels (0, 60, 100, and 150 g) were used. The experiment was done in two folds: the leachate and treated cocopeat examination for their chemical properties. The General Linear Model procedures were used for Analysis of Variance at ($P \leq 0.05$). The results showed that the addition of Ca(NO3)2 100 g extracted significantly more K and Na in the leachate than Ca(NO3)2 0.0 g and 60 g. The EC levels in the leachate increased with the application levels

of Ca(NO3)2 while the pH levels were reducing. In the treated cocopeat, Ca(NO3)2 100 g and soaking duration 36 hours significantly reduced K and Na and sufficiently supplemented Ca and N. Irrespective of Ca(NO3)2 and soaking durations, after the cocopeat is washed, the EC and pH values fall within their suitable ranges. There was a strong negative correlation between Ca and Na, Ca and K, and between Na and EC. Also, strong positive correlation between Ca and N and Ca and EC. Effective supplementation of Ca and N, and optimal reduction of K and Na by 78.44% and 92%, respectively can be achieved with 100 g of Ca(NO3)2 1.5 kg−1 of cocopeat in 15 liters of water with a soaking duration of 36 hours.

Conclusion

The untreated coco peat has high K, Na, and EC with limited Ca and N concentrations. Through the use of Ca $(NO_3)_2$, these excessive elements can be minimized while the limited elements are supplemented. In the leachate, Ca $(NO_3)_2$ is seen to have the ability for K and Na extraction while increasing the EC and reducing the pH. After the coco peat is soaked and washed, the K, Na, Ca, N, EC, and pH are more responsive to the main effects of $Ca(NO_3)_2$ and soaking duration than the interaction effects of $Ca(NO_3)_2$ × soaking duration. Comparatively, the single effect of Ca $(NO_3)_2$ extracted more K and Na compared to the single effect of soaking duration. The use of 100 g Ca $(NO_3)_2$ significantly extracts K and Na in coco peat. Although 150 g of Ca $(NO_3)_2$ extract Na by 22.99% more than 100 g of Ca $(NO_3)_2$, the minimal ranges are significantly the same. Calcium and N significantly increase with an increase in Ca $(NO_3)_2$ up to 150 g. After the coco peat is washed, the EC and pH reduce within their suitable ranges irrespective of Ca $(NO_3)_2$ levels or soaking durations. Soaking duration 36 hours is an equilibrium point for Ca and N supplementation, and for K and Na extraction. Higher Ca and N supplementation in coco peat is observed when 100 g of Ca $(NO_3)_2$ is used. On the other hand, K and Na tend to decrease faster in 0 g and 60 g compared to 100 g and 150 g of $Ca(NO_3)_2$. There is a strong negative correlation between Ca versus Na, Ca versus K, Na versus N, and between Naversus EC. Also, there is a strong positive correlation between Ca versus N, Ca versus EC, and between N versus EC. For optimal reduction of K by 78.44% and for effective supplementation of Ca and N in coco peat, 100 g of Ca $(NO_3)_2$ 1.5 kg^{-1} of coco peat in 15 litres of water with a soaking duration of 36 hours is very effective. Sodium is also significantly reduced by 95.83% when 150 g is used and by 92.59% when 100 g of Ca $(NO_3)_2$ with a constant soaking duration of 36 hours. Information about the leachability of K and Na is important for the standardization of coco peat for a variety of horticultural crops grown in a hydroponics system using coco peat as a growing medium.

6.0 Enhancing Growth and Yields of Seed Potato in farmers' fields (Seed bulking/Multiplication)

6.1 Effects of supplemental irrigation on yield, water use efficiency and nitrogen use efficiency of potato grown in Mollic Andosols

Citation: Satognon, F,. Lelei, J.J,. Owido, S.F.O. (2021). Effects of supplemental irrigation on yield, water use efficiency and nitrogen use efficiency of potato grown in Mollic Andosols. *Environmental Systems Research*. https://doi.org/10.1186/s40068-021-00242-4.

Abstract

The yield of potato (8.86 tonnes/ha), the second staple food and cash crop in Kenya has remained low due to a reduction in seasonal precipitation and low soil fertility. Drought or dry periods between rainfall seasons and increased temperatures, which leads to high crop evapotranspiration, are experienced in 70-80% of the smallholder farms. Among major elements require by potato, nitrogen is the most important influential element but it is deficient in most potato-growing soils in Kenya because of nutrient depletion with inadequate nutrient replenishment results from continuous production. Hence the introduction of supplemental irrigation with an adequate application of this nutrient could increase crop yields. Therefore, this study was conducted in Nakuru, one of the major potato growing areas in Kenya, to determine the effects of full supplemental irrigation (FI) and four nitrogen levels, N0(0), N1(60), N2(90) and N3(130 kg N/ha) on tuber yield, water use efficiency (WUE) and nitrogen use efficiency (NUE) of potato grown in a mollic Andosol in Kenya compared to conventional rain-fed potato production.

Conclusion

Yield components of potato were largely more responsive to the interaction effect of irrigation × N-fertilisation than a single effect of irrigation and N-fertilisation. The highest total tuber yield and marketable were obtained under FI with an application of 130 kg N/ha. In this study, WUE was high when applying 130 kg N/ha. The NUE consistently increased with the increase in N rate up to 90 kg/ha in the two irrigation treatments. There was a high positive correlation between tuber number/plant, tuber yield, WUE and NUE. This study highlights the importance of introducing supplemental irrigation in the potato production area in Kenya. Farmers can achieve a high potato yield and marketable tuber yield in Mollic Andosols when water deficits of the growing season are eliminated with supplemental irrigation and an application of 130 kg

N/ha. This study also recommends further research on water regimes and irrigation methods that can lead to high potato yield with water-saving in Mollic Andosols.

6.2 Use of Green Seeker and CM-100 as manual tools for nitrogen management and yield prediction in irrigated potato (Solanum tuberosum) production

Satognon, F., Lelei, J.J., Owido, S.F.O. (2021). Use of Green Seeker and CM-100 as manual tools for nitrogen management and yield prediction in irrigated potato (Solanum tuberosum) production. ***Archives of Agriculture and Environmental Science*****, 6,121-126. https://doi.org/10.26832/24566632.2021.060202/**

Abstract

This study evaluated the possibility of the use of GreenSeeker sensor and CM-100 chlorophyll content meter for in-season N and yield prediction in order to promote timely split N application in potato production in Kenya. Four N-fertilization rates; N0 (0), N1 (60), N2 (90) and N3 (130 kg N/ha) were led out in a Randomized Complete Block Design (RCBD) in a Greenhouse for two seasons. The results showed that % N leaf content was significantly affected by N rates. The % N leaf content and potato leaf chlorophyll content decreased as the season continued, whereas the Normalized Difference Vegetation Index (NDVI) increased as the season continued. CM-100 values were significantly correlated with % N leaf content at vegetative (r=0.86***) and tuber initiation (r=0.74***) growth stages of the crop whereas the NDVI values were only significantly correlated with % N leaf at tuber initiation (r=0.82***). A significant relationship was found between CM-100 values taken at different potato stages (end of vegetative, tuber initiation, bulking and maturation stages) and tuber yield (r=0.90***, 0.82***, 0.47* and 0.41*). The NDVI values at end of vegetative growth, tuber initiation and maturation of potato were also significantly correlated with tuber yield (r=0.81***, 0.43* and 0.54*), except at bulking stage (r=0.33). For efficient in-season N management and yield prediction, CM-100 and GreenSeeker are recommended at an early stage of the crop. Further research in the different potato growing areas in Kenya to establish the different thresholds at different potato growth stages is recommended.

Conclusion

To achieve an optimum potato tuber yield and high quality, N fertilization should not only be

proportional to the crop need but be applied at right time. Hence, split application of mineral N fertilizer throughout the growing season is the suitable and appropriate approach to match potato N requirement and supply. This study found that CM-100 can be used for in-season N management at vegetative and tuber initiation stages whereas GreenSeeker sensors can only be used at the vegetative stage due to its early saturation. For yield prediction, CM-100 can be used throughout the growing season of potato. The GreenSeeker sensor can also be used for tuber yield prediction at all stages of potato except the tuber bulking stage. For efficient yield and nitrogen recommendation, CM-100 is recommended but GreenSeeker can also be used at an early stage of the crop. The study recommends the development of a manual remote sensor for late N management in potatoes. This study also recommends further investigations in the subject in the different potato growing areas of Kenya to establish the different thresholds at different stages of the crop.

Post-Harvest Handling

7.1 Effects of different light exposure regimes on pre sprouting and yield performance on potato (Solanum tuberosum L.) seed in Central Rift, Kenya

Moletsane, M. S., Kimurto, P.K. & Oyoo, M. E. (2021). Effects of different light exposure regimes on pre-sprouting and yield performance on potato (Solanum tuberosum L.) seed in Central Rift, Kenya. ***African Journal of Agricultural Research*****, 17(7), 1039-1048.**

Abstract

The major problems affecting potato (Solanum tuberosum L.) production in Kenya include timely availability of well-sprouted seed potato tubers. Appropriate techniques of availing well-sprouted potato seed for timely planting for different potato varieties with varied seed dormancy periods is one of the major challenges for yield increase. A study was conducted to determine the effects of different light exposure regimes on pre-sprouting, seed germination and yield performance of potato in Central Rift, Kenya. Three light exposure regimes of 8, 12, and 16 h were evaluated in lighthouse for their effects on breaking dormancy and enhancing pre-sprouting in three potato varieties. The lighthouse experiment was laid on complete randomized design (CRD) with three replicates while field experiment was laid on randomized complete block design (RCBD). Data collected on the number of sprouts, sprout lengths, diameters and colour; emergence rate, plant height, number of tubers per plant, and tuber weight were subjected to analysis of variance (ANOVA) and means were compared using Duncan multiple range test ($P<0.05$). There were significant differences ($P<0.05$) in different light exposures on traits. Light exposure regimes significantly increased sprout thickness by 31 to 43% and sprout length by 15 to 39% when compared with those of control treatment. Light exposure of 16 h resulted in the highest percent seed germination and tuber weight by 9.5 and 25%, respectively. This suggests that increasing the period of light exposure by using a simple fluorescence system in pre-sprouting stores will enhance breaking of potato seed dormancy resulting in increased yields.

Conclusion and Recommendations

Light exposures significantly enhanced the quality of sprout by influencing the thickness and length. This was highly correlated to growth and yield of potato plants. Seeds tubers with good quality sprouts in terms of length and thickness can lead to high vigorous plants which will give good yield. Light exposures reduced the sprouting time by 23 days (from 44 to 21 days)

which is advantageous in increasing growing cycles per annum. It also significantly increased sprout thickness and sprout length. Light exposure of 16 h resulted in the highest percent plant germination. The best light regime was 16 h which increased tuber weight by 25% as compared to 8 h that increased tuber weight by 12%. The 16 h of light attained the highest yield in Shangi and Kenya Karibu. These varieties could be recommended when the suitable pre-sprouting conditions are available to the farmers.

8.0 Crop Protection

8.1 Effects of Seed Tuber Size of Potato Varieties on Fungicide Spray Regime, Weed Infestation and Net Farm Income in Potato Production

Kilonzi, J. M., Githui, D., Pwaipwai, P., Kawira, C., Otieno, S., Kelele, J., Ng'ang'a, N., Nyongesa, M., Mafurah, J., & Kibe, A. (2024). Effects of seed tuber size of potato varieties on fungicide spray regime, weed infestation and net farm income in potato production. ***Potato Research, 67,*** **23–47. https://doi.org/10.1007/s11540-024-09708-1**

Abstract

Field studies were conducted to determine the contribution of seed tuber size on late blight management, weed abundance, crop performance and net farm income. Seed tuber sizes were as follows: small size (15 to 27 mm), size 1 (28–35 mm), size 2 (36–45 mm) and size 3 (46–60 mm) of Shangi, Kenya Mpya, Unica and Dutch Robijn potato varieties. Fungicide spray regimes were weekly, biweekly and triweekly. Data on late blight severity, weed frequency and density, growth parameters, costs and revenues were collected. Results revealed that seed tuber size, variety and fungicide application regime had significant ($P \leq 0.05$) effects on late blight severity. As such, weekly spray interval reduced blight severity by 50%, and the blight-tolerant variety (K. Mpya) suppressed the disease to below 1%. Whereas the lowest disease severity was observed on seed sizes 2 (44%) and 3 (43%), the highest blight score was recorded in small seed size (59%) after 70 days of emergence. Notwithstanding the variety used, a combination of seed size 2 with weekly spray interval showed the lowest disease progression as compared to any other combination. Crop growth parameters differences were phenomenal among seed sizes: stem count increased six-fold, while height and canopy were double the observation made in the small seed size for seed sizes 2 and 3. Weed frequency, relative frequency and density decreased with increasing seed size planted. As a result, seed tuber sizes 1, and 3 augmented marketable yield by 49%, 62% and 65% as compared to the small tube size, respectively.

However, seed size 2 had the highest cost–benefit ratio (1.50), followed by size 3 (1.05).

Conclusion

Kenya Mpya variety reduced late blight severity by up to 100%. Weekly spray intervals provided the highest level of protection against late blight as compared to other regimes. Seed size had a significant contribution to crop growth, blight infection, weed management, dry matter, number of tubers per hill, seed distribution and yield formation but less effect on root biomass index. As such, seed size 3 reduced fungicide application frequency by 25%. The highest growth vigour, tuber count and yield were observed in seed size 3. Minimal weed species and weeding frequency were observed in plots in which seed size 3 was planted. Additional weeding and fungicide application frequency was required for plots with small size and seed size 1. However, the highest net benefits were observed in seed size 2. Unica had the highest yield as compared to Shangi, K. Mpya and Dutch Robijn.

8.2 Efficacy of *Trichoderma asperellum* Seed Treatment and Ridomil® Application in Managing Late Blight on Potato

Mutuku, K. J., Joseph, M. J., Wabomba, N. M., & Mwangi, K. A. (2021). Efficacy of Trichoderma asperellum seed treatment and Ridomil® application in managing late blight on potato. *World Journal of Agricultural Research*, *9*(2), 42–52. https://doi.org/10.12691/wjar-9-2-1

Abstract

Potato seed tubers latently infected with *Phytophthora infestans* initiate late blight that requires early fungicide application raising economic and human concerns. The objective of the study was to determine the efficacy of *Trichoderma asperellum* seed treatment and Ridomil® (Metalaxyl 4% and Mancozeb 64%) application to manage late blight. Ridomil® was applied at 21-, 14- and 7-day intervals on seed tuber and apical cuttings pre-treated with *T. asperellum* at 33 % (3 × 106), 66% (7 × 106) and 100% (1 × 107 CFU/mL) concentration by either dipping or injection. Results revealed that 7- and 14-day spray intervals were not significantly different (P=0.05) in terms of yield and late blight severity. Rooted apical cuttings had 7.4% higher disease severity resulting in 2.3% lower yield than crop from seed tubers. *T. asperellum* at 66% and 100% concentrations reduced disease severity by 26% and 27% respectively. Pericardial injection had 8.3% higher yield and conversely 7.8% higher disease severity than dipping. The combination of *T. asperellum* at 66% concentration with a 14-day spray interval provided better late blight management. The results suggest that seed treatment by dipping using 66% *T. asperellum* suspension could increase fungicide application interval by 7 days while improving on yield.

Conclusion

The findings of this study support the following conclusions; First, the rapid establishment and quick canopy development of rooted apical cuttings used as seed provides an opportunity for the crops to escape blight attack, more so in the short rain season than in the long rain season. Secondly, effective blight suppression through seed treatment is achieved at optimal *T. asperellum* suspension concentrations of 66% and 100%. Thirdly, there is synergistic benefit of combining *T. asperellum* optimal concentrations with 7-or 14-day spray interval with metalaxyl based fungicide to control late blight severity. Lastly, we conclude that seed treatment through dipping could be used prior to planting with subsequent fungicide usage on emerging crops. This approach reverses risks of latently infected yet symptomless seed tubers that are a common source of primary *P. infestans* inoculum responsible for late blight epidemics early in

the cropping season. Future studies could be conducted to establish possible roles and mechanisms of *T. asperellum* in potato growth and yield promotion.

8.3 Ex ante economic impact assessment of the 3R-gene potato in Kenya

Kihiu, E., Ghislain, M., Kibe, A. M., Nancy, N., Gatto, M., & Falck-Zepeda, J. B. (2025). Ex ante economic impact assessment of the 3R-gene potato in Kenya. *PLoS ONE, 20*(3), e0309329. https://doi.org/10.1371/journal.pone.0309329

Abstract

Potato late-blight disease is a major constraint to potato production in Kenya. The use of fungicides to control the disease is limited by the practice of delaying application until the symptoms are visible, and inappropriate application rates and methods. Biotech crops, such as 3R-gene potato, are providing sustainable solutions to crop protection challenges in agriculture, but little is known about their social and economic potential in any country. To gain greater insights into the potential of 3R-gene potato in Kenya, this study evaluates the economic benefits of Asante, Shangi, and Tigoni potato varieties, which are resistant to late blight. Data from experts along the potato value chain and secondary sources are analyzed using the economic surplus model and a real options model. First, experts' opinions revealed that late-blight disease is responsible for 23% of production loss annually, and that 12% of the production costs are due to the use of fungicides to control late blight disease. Secondly, the study results suggest that the release of 3R-gene Shangi would generate the greatest economic benefits of KES 845.9 million (US$ 8.2 million) annually. The expected net benefits of 3R gene Asante are KES 7.3 million (US$ 0.07 million) annually. 3R-gene Tigoni, with the lowest potential adoption rates, would be expected to realize a negative net of KES of -1.26 million (US$ -0.01 million) annually. Significant potential economic gains, which is expected to increase with better awareness of biotech crops, support the immediate release of 3R-gene Shangi in Kenya.

Conclusion

The result of this study supports the findings of ex ante economic impact assessment studies on economic benefits of biotech crops in Africa. The findings suggest high returns with the adoption of 3R-gene Shangi, which is estimated to have the highest potential adoption rate based on experts' experience of the variety's superior qualities relative to 3R gene Asante and 3R-gene Tigoni. The willingness to adopt 3R-gene Shangi, and in turn the expected higher

benefits are attributed to its resistance to late blight and the excellent traits of the non-3R-gene, which are retained in the transformed variety. Similar observations are made in where the benefits of agricultural biotechnology are greater with larger adoption areas. The changes in producer surplus are approximately twice as large as the consumer surplus. The distribution of benefits is similar to that in agricultural studies in which the demand elasticity (in absolute terms) is larger than the supply elasticity. We note that while producers are expected to benefit less in a closed economy from an increased supply that tends to suppress the domestic price relative to a small open economy, the increase in potato demand in the Kenyan case is likely to suppress a downward price change. To put the potential benefits of the 3R-gene potato in relatable terms, our calculations suggest that 3,063 people will be lifted out of poverty with the adoption of 3R-gene Shangi annually, with a total of 91,884 individuals escaping poverty during the expected lifetime of the 3R-gene Shangi cultivation period. This reflects the potential economic power of the 3R-gene potato.

It is important to note the limitations of ESM when interpreting the results. The economic surplus model is sensitive to changes in key assumptions, mainly estimated yield and cost changes, and elasticities. In addition, while yield and cost changes are based on average values at the regional level, the assumed changes do not consider heterogeneity among farmers in the region. This paper has attempted to overcome these shortcomings through sensitivity analysis, using various scenarios, and assumptions informed by an extensive literature review, national statistics and parameters, and expert views. In the scenarios evaluated, the investigated effects of a delay in approval shows the greatest negative impact on the potential benefits of adopting the 3R-gene potato. Research, development, and regulatory delays reduce the benefits by 36%, but do not alter the importance and positive impact of 3R-gene Shangi. It is therefore prudent that regulatory efficiency and effectiveness be enhanced to realize the maximum benefits of the technology to Kenya's producers and consumers.

Apart from sensitivity to key parameters, the modeled changes in the supply within the ESM does not take into account the transactional costs that are likely to occur with the release of agricultural technologies. Transactional costs can be included after the supply shift impacts have accrued, reducing the level of benefits. Transaction costs such as those associated with dissemination, promotion, and adoption of the 3R-gene potato have not been considered here, given that they are not part of the R&D process of the not-for-profit and local developing institutions involved, and we continue to be interested in the returns on R&D investment to all developers. However, to mitigate the possibility of benefit overestimation, conservative assumptions about R&D costs and cost changes were made, in line with similar studies. A

robustness check of the estimated ESM benefits was performed using the real options model. Although no plausible risk of cultivating and consuming 3R-gene potato varieties has been identified, the real options model allows us to evaluate the economic benefits of the 3R-gene potato under conditions of irreversibility and uncertainty regarding their future benefits and costs.

The 1.271 hurdle rates calculated using the real options approach indicate that 1KES of incremental irreversibility costs needs to be compensated by about KES 1.271 of social incremental reversible and irreversible benefits. Furthermore, the results indicate that if Kenya delays the release of 3R-gene Shangi, the country would be losing additional benefits of about KES 13.54 billion (US$ 131.11 million) annually. Our estimates of the potential economic benefits of 3R-gene might have underestimated the potential benefits by ignoring the indirect benefits of the technology. Nevertheless, with potato LB disease remaining a severe threat to potato performance in Kenya, the results show its adoption has great potential to improve consumer and producer surplus, greatly affecting food security and the profitability of smallholder farmers. Kenyan experts, including those who specialize in potato production, anticipate that outreach, communication and other awareness programs on the potential benefits of the 3R-gene potato are likely to stimulate higher adoption rates and increase benefits to producers and consumers in Kenya, especially with a widely disseminated and damaging disease such as late blight. In general, these findings provide a significant step towards the release, dissemination and adoption of 3R-gene potato in Kenya. The results substantially improve our understanding of the potential economic performance of 3R-gene potato in Kenya and will help ensure that the enabling environment in the country continues to be conducive to valuable and safe technologies for Kenyan farmers to improve their food security.



8.4 Influence of Plant Health Clinic Diagnostic Services on Potato Production among Smallholder Farmers of Molo Sub-County, Kenya

Chepkoech, B., Maina, S.M. & Ng'eno, J.K. (2023). Influence of Plant Health Clinic Diagnostic Services on Potato Production among Smallholder Farmers of Molo Sub-County, Kenya. *Journal of Agricultural Extension and Rural Development*, 15(1), 13-21

Abstract

As compared to other African countries, Kenya continues to depend on agriculture as source of food and for economic development. Horticulture forms the bulk of agriculture with potato being ranked first in the vegetable category in terms of production and value. It is a lucrative cash and food crop grown in many parts of Kenya. Molo Sub County being one of the regions where the crop is grown enjoys economic value in terms of income generation and food provision. Given the importance of the crop, agricultural extension such as plant health clinic training services has a key role in enhancing production through training farmers in innovations on potato protection measures and production improvement. This study thus established the influence of plant health clinic training services on potato production among smallholder farmers in Molo Sub-County, Kenya. Through simple random sampling techniques and purposive sampling, 152 smallholder potato farmers and 10 key informants respectively were selected for the study. Data were collected using structured questionnaires and interviews. The data was analyzed using descriptive statistics and inferential statistics. The type of training access, the relevance of training services, frequency of receiving training, modes of training, and use of training services were found as significant predictors variable influencing potato production at 0.05 significant level ($p = 0.000$, $p = 0.000$, $p = 0.009$, $p = 0.000$ and $p = 0.000$, respectively). **The study recommends that adequate plant health clinic training services should be provided in potato production.**

Conclusions

Conclusively, frequency of diagnosis from plant health clinics, perception of the role of diagnostic services and use of diagnostic services empowers potato production of smallholder farmers. Therefore, it becomes important to enhance use of diagnostic services, perception of the role of diagnostic services and frequency of plant health clinics diagnostic services among smallholder potato farmers. **It is important to empower smallholder potato farmers with**

the latest know-how and innovations on production technology and information technology through these services. Smallholder potato farmers should **be sensitized to utilize plant health clinic diagnostic services** in potato production to improve production through enhancement of pests and disease resilience.



9.0 Gender and Development studies

9.1 Gender Disparities Influence on Adoption of Climate Smart Technologies in Seed Potato Production in Ol-Kalau Sub County Kenya

Journal: International Journal of Social Sciences and Information Technology

Citation: Wamaitha, T.B., Rotich, C.L. & Kibe, A.M. (2022). Gender Disparities Influence on Adoption of Climate Smart Technologies in Seed Potato Production in Ol-Kalau Sub County Kenya. *International Journal of Social Sciences and Information Technology*, 8:6.

Abstract

Climate Smart Agriculture (CSA) offers farmers' the opportunity to improve their economic and food security status. However, farmers' success in adoption of CSA technologies is constrained by gender disparities, i.e.,

i. level of education,
ii. access to knowledge in CSA, (e.g. crop rotation, seed sizews, varieties, fertilizers,
iii. access to finances, availability
iv. nature of CSA technologies,
v. **cultural norms and values,**
vi. **attitude of farmers towards the**
vii. **CSA, access**
viii. **lack of CSA skills.**

This study therefore aimed at determining the variance in adoption of CSA technologies in seed potato production as influenced by gender disparities in Ol-kalau sub-county of Nyandarua county, Kenya. The study results indicated that with the exception of attitude of farmers' there were significant gender differences in adoption of CSA technologies, with males dominating in all CSA practices. Financial services are major reasons why most farmers have not adopted CSA practices. In order to enhance female farmers, women should be encouraged to participate in ventures that help them access, generate and mobilize resources so that they can acquire credit; be supported to access relevant CSA knowledge, skills and resources. Thus, they will be able to offer supplementary support to their male counterparts in adoption of seed potato CSA

farming technologies.

Conclusions

From the study results, the following conclusions have been drawn: The study set out to evaluate the disparities among men and women in accessing necessary resource and how this **allocation influences adoption of CSA practices in seed potato production** in Kenya, and important potato growing country with challenges in seed production. Access to finances had a remarkable effect on adoption of CSA technologies among seed potato farmers this was so because the adoption of CSA comes with a **cost in terms of purchasing the technologies**.

In addition, cultural norms and values affected female farmers significantly reducing their participation in seed potato production.

Access to knowledge and skills on CSA can be successfully achieved through trainings and seminars where the females are able to attend and get enlighten on different CSA technologies to adopt as far as seed potato production is concerned.

Attitude of other farmers did not affect any gender as most of seed potato farmers are convinced on importance of seed potato production.

Gender inequalities in social economic factors tend to favor men than women in most cases.

This limit women access to resources causing a big barrier to women's adoption of new technologies, which are needed in order to practice what the technologies demand to be done by the farmers.

Recommendations

The study recommends that; men and women farmers should be encouraged to participate in ventures that help them access, generate and mobilize resources so that they can acquire credit; be supported to access relevant CSA knowledge, skills and land. Cultural norms and values affecting women participation in seed potato production should be put into consideration during implementation and interventions. Gender roles should not be assigned according to cultural stereotyping and biological roles. In order to create time for female farmers to participate in agriculturally based training, workshops and seminars in CSA seed potato production, women should be prioritized in access to knowledge and training on CSA skills, access to finance that will enable them access new technology necessary for seed potato production. Greater workload with gender roles reduces female participation in adoption of CSA seed potato production. Time saving technologies for domestic chores (cleaning utensils, washing clothes and cooking) would act as best solution in creating the much-needed time required for women

participation in seed potato production, leading to higher yields and incomes in rural households.

Policy

9.2 Does access to credit influence the adoption of good agricultural practices? The case of smallholder potato farmers in Molo Sub-County, Kenya

Journal: Journal of Agricultural Extension and Rural Development

Citation: Masca, U., Kyule N. M. & Bor. E.K. (2021). Does access to credit influence the adoption of good agricultural practices? The case of smallholder potato farmers in Molo Sub-County, Kenya. *Journal of Agricultural Extension and Rural Development*, 14(1), 24-32

Abstract

Potato (*Solanum tuberosum* L) is the major food and cash crop for many rural and semi-urban people in Kenya. It contributes to the national food security, nutrition, and income generation for those who are involved in its value chain. Despite its importance, smallholder potato farmers in Molo sub-county are still recording low potato yields which could partly be attributed to the low adoption of Good Agricultural Practices (GAPs). The main aim of this study was to investigate the influence of access to credit on the adoption of GAPs among smallholder potato farmers in Molo sub-county, Kenya. The research employed a descriptive survey design. A sample of 108 smallholder potato farmers from four wards: Molo, Turi, Elburgon and Marioshoni in Molo sub-county was selected using a purposive, random sampling technique. Data analysis was done using descriptive statistics and a binary logistic regression model was used to test the hypothesis at 5% level of significance. Statistical Package for Social Science (SPSS version 22) was used in data analysis. Therefore, the null hypothesis that access to credit has no statistically significant influence on the adoption of GAPs among smallholder potato farmers was rejected. The findings of the study were that access to credit significantly ($p<0.05$) influences the adoption of GAPs among smallholder potato farmers in Molo sub-county. This study recommends that the government of Kenya should facilitate the lowering of the rate of interest on access to credit among smallholder potato farmers, to make it easier for them to have access to credit

Conclusion and Recommendations

The findings showed that there is low access to credit among the smallholder potato farmers due to the high interest rate and long credit procedure process. The binary logistic regression model showed that access to the credit had a positive statistically significant influence on the

adoption of GAPs among smallholder potato farmers since p-value was less than 0.05. Additionally, the major limitation for this study was language barrier and it was overcome by hiring a translator to help the researcher to easily communicate with the farmers during data collection. This study recommends that the government should facilitate lowering the rate of interest on accessing credit among smallholder potato farmers, to make it easier for them to access credit from the Banks, SACCOs, and microfinance institutions.

9.3 Producer Needs and Collaborative Opportunities For Enhancing Potato Productivity In Nakuru Kenya

World Potato Congress 2026

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Abstract

The potato (*Solanum tuberosum L.*) stands as Kenya's second most important staple crop, playing a pivotal role in national food security, rural incomes, and the country's agro-industrial growth. With over 161,000 hectares under cultivation, the sector produces approximately 1.5 million tonnes annually, contributing an estimated KSh 50–60 billion to the economy. Beyond its fiscal impact, the value chain provides direct livelihoods for more than 800,000 farmers and supports an additional 2.5 million people indirectly. Nakuru County remains at the heart of this industry, accounting for over 60% of national production and hosting more than 50,000 potato farmers. A cross-sectional descriptive survey was executed using a mixed-methods approach. Multi-stage stratified random sampling was conducted across five sub-counties: Bahati (n=101), Kuresoi North (n=138), Kuresoi South (n=126), Molo (n=139), and Njoro (n=128). The sample size of n=632 smallholders was validated using the Nassiuma (2000) formula at a 28% coefficient of variation and a 2% margin of error. Quantitative data were analyzed via SPSS (v25) using chi-square (X^2) tests of independence and descriptive metrics, while qualitative thematic nodes were evaluated via NVIVO (v12). The assessment revealed near gender parity in production roles, with 52.2% male and 47.8% female farm managers (X^2 = 1.23, $p > 0.05$). The study revealed that only 23.9% of potato farm managers in the region are youth (under 35 years of age), with 76.1% of the active farmers being above 35 years. However, access to critical extension support was profoundly unequal, with 57.8% of all smallholders cut off from advisory services ($p < 0.01$). Technical gaps were severe: only 22.3% utilized certified seeds , while high biotic pressures were reported for Bacterial Wilt (83.7%) and Late Blight (83.0%).

While fertilizer use at planting was nearly universal (95.3%), application rates were statistically unoptimized due to a lack of rapid soil testing. Postharvest vulnerabilities were widespread, as 78.0% of farmers lacked formal infrastructure, suffering mean volume losses of 16.3%. Furthermore, market dynamics were deeply asymmetrical; 91.6% sold at the farm gate , where brokers unilaterally dictated peak-season pricing for 81.6% of producers ($p < 0.001$). Consequently, baseline adoption of CSA technologies remains critically low at just 3.5%. The smallholder potato value chain in Nakuru County is bound to a low-input, low-output equilibrium enforced by structural market gaps, asymmetric resource access, and systemic institutional exclusion. Therefore, transitioning the value chain requires immediate deployment of Climate-Smart Agriculture (CSA) frameworks. Specifically, scaling up gender-inclusive training, institutionalizing rapid soil testing, and setting up localized Storage networks are critical steps to eliminate the yield gap and build smallholder climate resilience.

Keywords: **Potato Value Chain, Smallholder Productivity, Postharvest Management, Climate-Smart Agriculture, Gender**

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